Supplementary Information

Extreme-by-design urban lagoon: trophic simplification and non-random functional organization under chronic stress

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Appendix S1. Study area and water quality characterization

The study site is a small temperate urban lagoon located in the Mesopotamian Pampas, within a multipurpose recreational park in the city of Gualeguaychú (Argentina) (**Figure S1**). Briefly, the lagoon is shallow, floodplain-associated, and seasonally inhabited by a diverse bird community, including migratory herons. According to Crettaz-Minaglia & Gianello (2023), the system exhibits typical features of shallow, eutrophic ecosystems, with low spatial heterogeneity and marked seasonal dynamics. Water temperature closely follows air temperature, while dissolved oxygen levels are highly variable and can drop below 2 mg/L, indicating hypoxic events. Electrical conductivity and pH remain relatively stable, whereas water transparency is consistently low. Nutrient concentrations are elevated year-round, with frequent peaks in soluble reactive phosphorus, total phosphorus, and chlorophyll-a, confirming the lagoon's hypereutrophic status. Microbiological indicators (mesophilic aerobic bacteria and total coliforms) show high counts during several sampling events, consistent with high organic loads.

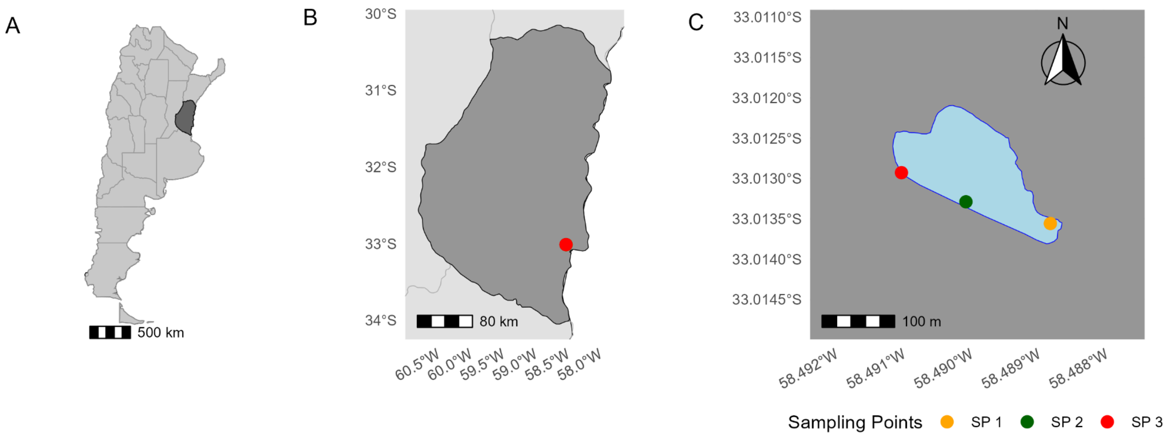

Figure S1. Study area corresponding to the lagoon of the Parque Unzué (Gualeguaychú., Entre Ríos, Argentina). A) Map of Argentina with the province of Entre Ríos highlighted in dark gray. B) Enlarged map of Entre Ríos showing the location of the sampling area. C) Laguna del Parque Unzué with its respective sampling points. This figure was produced using R software (version 4.5.1).

Appendix S2. Samples collection

Between May 2015 and November 2019, surface water and sediment samples were periodically collected from the lagoon at Parque Unzué. Qualitative microphytoplankton (20–200 μm) and zooplankton (> 50 μm) samples were obtained using a 15 μm mesh Zeppelin net and fixed with 50 % Transeau solution (Gianello et al., 2019). Samples were examined under an Olympus® microscope at 40–100 × magnification. Microphytoplankton were identified to genus level using taxonomic keys (Komárek & Komarkova, 2003; Shubert, 2003; Gerroth, 2003; Stoermer & Julius, 2003; Kingstom, 2003; Kociolek & Spauding, 2003 a, b), while zooplankton were classified into nauplii, rotifers, copepods, and cladocerans. Sediment samples were collected with an Ekman dredge and fixed in 10 % formaldehyde for macroinvertebrate identification (Crettaz-Minaglia et al., 2018). In the laboratory, samples were examined under a Motic® stereomicroscope at 10–30 × magnification, and macroinvertebrates (> 250 μm) were identified to family level using appropriate dichotomous keys (Crettaz-Minaglia et al., 2018). A list of fish and aquatic reptile species observed at the site was obtained from Gianello (2016).

Appendix S3. Spatial and temporal food-web construction and analysis

Structural and functional food webs were constructed and analyzed to capture both spatial and temporal dimensions of community organization. The same analytical workflow described below was applied (i) to a spatially integrated network, combining all taxa and trophic interactions recorded between 2015 and 2019, and (ii) to temporally resolved networks, one for each sampling date (n = 16), to assess seasonal and interannual variability.

Food webs were analyzed using Python within the Google Colab environment. Each identified organism or taxonomic group was assigned to a node to construct the structural network. Nodes were connected to one or more resources or prey based on bibliographic records of species interactions in Neotropical freshwater communities. A linked list structure was used to organize the data, with each row recording a consumer and its corresponding resource.

A functional network was also constructed by assigning each node to a functional category based on its trophic role or broader ecological traits. Microphytoplankton were classified as autotrophic, mixotrophic, or facultatively mixotrophic (Reynolds et al., 2002; Padisák et al., 2009). Macroinvertebrates were grouped into predators, gathering collectors, filtering collectors, scrapers, shredders, and omnivores (Cummins et al., 2005; Rodríguez-Barrios et al., 2011). Zooplankton were categorized as consumers, and fish and reptiles as omnivores, detritivores, or carnivores. Conceptual nodes representing basal resources and microbial compartments that were not directly sampled but are essential components of the trophic structure were also incorporated, including dissolved organic matter (DOM), fine and coarse particulate organic matter (FPOM, CPOM), periphyton, and bacterioplankton.

The following topological metrics were quantified for both networks: number of nodes and links, connectance, average degree, average and maximum trophic level, modularity, mean degree centrality, mean betweenness centrality, trophic chain length (TCL), and the proportion of links from basal sources.

For the functional network, redundancy and the domain of functional pathways were additionally calculated. Functional redundancy was defined as the ratio between the total number of taxa and the number of distinct functional categories, providing a measure of the average number of taxa performing similar roles within the network. The conceptual and analytical framework follows Polazzo et al. (2022).

To assess whether temporal observed network patterns differed from random expectations, we compared each empirical network with directed degree-preserving null models. Null networks preserved the number of nodes and links as well as the in-degree and out-degree sequence of each node, while randomizing the identity of trophic connections through a stub-matching rewiring procedure. This approach provides a baseline for the structure expected by chance, against which we evaluated our empirical values. Metrics that did not differ significantly from the null distribution were interpreted as consistent with structural simplification (supporting H3). In contrast, metrics that deviated from null expectations (e.g., higher modularity in functional networks) were interpreted as evidence of functional organization beyond random assembly (supporting H1). For each empirical network corresponding to the sampling date, we generated 1000 randomized networks by rewiring trophic links while preserving each node's in- and out-degree sequence (stub–matching procedure). For each randomized replicate, we recalculated mean betweenness centrality and trophic chain length (TCL); modularity was estimated on a reduced set of 300 replicates, given its higher computational demand. Observed values were then compared against the corresponding null distributions, and standardized effect sizes were computed as z-scores:

$$z = (X_{obs} - \mu_{null}) / \sigma_{null}$$

where X_{obs} is the observed metric, and μ_{null} and σ_{null} are the mean and standard deviation of the null replicates, respectively. Empirical two–sided p-values were obtained as the proportion of null replicates with deviations from the null mean greater than or equal to the observed deviation. This approach allowed us to assess whether network modularity, betweenness, and TCL differed significantly from random expectations given the degree structure of the networks.

Table S1. List of the microphytoplankton of the Parque Unzué lagoon registered during the study period.

Phylum	Class	Order	Family	Genus	FG
Chlorophyta	Chlorophyceae	Sphaeropleales	Hydrodictyaceae	*Pediastrum*	J
Chlorophyta	Trebouxiophyceae	Chlorellales	Chlorellaceae	*Chlorella*	X1
Chlorophyta	Chlorophyceae	Sphaeropleales	Scenedesmaceae	*Scenedesmus*	J
Chlorophyta	Chlorophyceae	Sphaeropleales	Scenedesmaceae	*Coelastrum*	J
Charophyta	Zygnematophyceae	Desmidiales	Closteriaceae	*Closterium*	P
Charophyta	Trebouxiophyceae	Chlorellales	Chlorellaceae	*Dictyosphaerium*	F
Charophyta	Zygnematophyceae	Desmidiales	Desmidiaceae	*Staurastrum*	P
Chlorophyta	Chlorophyceae	Chlamydomonadales	Volvocaceae	*Eudorina*	G
Chlorophyta	Chlorophyceae	Sphaeropleales	Scenedesmaceae	*Tetrastrum*	J
Chlorophyta	Chlorophyceae	Sphaeropleales	Schroederiaceae	*Schroederia*	X3
Chlorophyta	Chlorophyceae	Sphaeropleales	Hydrodictyaceae	*Tetraedron*	G
Chlorophyta	Chlorophyceae	Chlamydomonadales	Sphaerocystidaceae	*Sphaerocystis*	F
Charophyta	Zygnematophyceae	Desmidiales	Desmidiaceae	*Staurodesmus*	N
Chlorophyta	Chlorophyceae	Sphaeropleales	Characiaceae	*Ankyra*	X1
Charophyta	Trebouxiophyceae	Chlorococcales	Scenedesmaceae	*Crucigenia*	J
Charophyta	Zygnematophyceae	Zygnematales	Zygnemataceae	*Mougeotia*	T

Charophyta	Zygnematophyceae	Desmidiales	Zygnemataceae	*Cosmarium*	N
Euglenophyta	Euglenophyceae	Euglenida	Phacaceae	*Phacus*	W1
Euglenophyta	Euglenophyceae	Euglenida	Phacaceae	*Euglena*	W1
Euglenophyta	Euglenophyceae	Euglenida	Euglenidae	*Trachelomonas*	W2
Euglenophyta	Euglenophyceae	Euglenida	Euglenidae	*Strombomonas*	W2
Euglenophyta	Euglenophyceae	Euglenida	Phacaceae	*Lepocinclis*	W1
Euglenophyta	Euglenophyceae	Euglenida	Euglenaceae	*Colacium*	
Bacillariophyta	Bacillariophyceae	Naviculales	Naviculaceae	*Navicula*	MP
Bacillariophyta	Bacillariophyceae	Cymbellales	Gomphonemataceae	*Gomphonema*	MP
Bacillariophyta	Bacillariophyceae	Naviculales	Amphipleuraceae	*Amphipleura*	
Bacillariophyta	Bacillariophyceae	Fragilariales	Fragilariaceae	*Fragilaria*	P
Bacillariophyta	Bacillariophyceae	Bacillariales	Bacillariaceae	*Nitzschia*	D
Bacillariophyta	Bacillariophyceae	Naviculales	Sellaphoraceae	*Sellaphora*	
Bacillariophyta	Bacillariophyceae	Naviculales	Pinnulariaceae	*Pinnularia*	
Bacillariophyta	Bacillariophyceae	Aulacoseirales	Aulacoseiraceae	*Aulacoseira*	B
Bacillariophyta	Bacillariophyceae	Cymbellales	Cymbellales	*Reimeria*	
Bacillariophyta	Bacillariophyceae	Cymbellales	Cymbellaceae	*Cymbella*	MP
Bacillariophyta	Bacillariophyceae	Cymbellales	Gomphonemataceae	*Gomphoneis*	
Bacillariophyta	Bacillariophyceae	Bacillariales	Bacillariaceae	*Hantzschia*	
Bacillariophyta	Bacillariophyceae	Cymbellales	Cymbellaceae	*Geissleria*	
Bacillariophyta	Bacillariophyceae	Surirellales	Surirellaceae	*Surirella*	MP
Bacillariophyta	Bacillariophyceae	Rhabdonematales	Tabellariaceae	*Diatoma*	D
Bacillariophyta	Bacillariophyceae	Thalassiosirales	Stephanodiscaceae	*Cyclotella*	A
Bacillariophyta	Bacillariophyceae	Naviculales	Naviculaceae	*Caloneis*	
Bacillariophyta	Bacillariophyceae	Naviculales	Pleurosigmataceae	*Gyrosigma*	
Bacillariophyta	Bacillariophyceae	Naviculales	Diploneidaceae	*Diploneis*	
Dinophyta	Dinophyceae	Gonvaulacales	Ceratiaceae	*Ceratium*	LM
Dinophyta	Dinophyceae	Gymnodiniales	Gymnodiniaceae	*Gymnodinium*	Lo
Cyanobacteria	Cyanophyceae	Spirulinales	Spirulinaceae	*Spirulina*	S2
Cyanobacteria	Cyanophyceae	Synechococcales	Merismopediaceae	*Merismopedia*	Lo
Cyanobacteria	Cyanophyceae	Synechococcales	Merismopedriacear	*Aphanocapsa*	K
Cyanobacteria	Cyanophyceae	Chroococcales	Choococcaceae	*Chroococcus*	
Cyanobacteria	Cyanophyceae	Nostocales	Aphanizomenonaceae	*Dolichospermum*	H1
Cyanobacteria	Cyanophyceae	Synechococcales	Pseudanabaenaceae	*Pseudanabaena*	S1

Cyanobacteria	Cyanophyceae	Chroococcales	Microcystaceae	*Microcystis*	M

Table S2. List of the benthic macroinvertebrates of the Parque Unzué lagoon registered during the study period. FFGs are the functional feeding groups classified as P = predator, FC = filtering collector, GC = gathering collector, S = scraper, O = omnivorous.

Phylum	Class	Order	Family	Genus	FFG
Annelida	Clitellata	Hirudinea			P
Annelida	Clitellata	Oligochaeta			GC
Mollusca	Gastropoda	Architaenioglossa	Ampullariidae	*Pomacea*	S
Mollusca	Bivalvia	Sphaeriidae	Sphaeriidae		FC
Mollusca	Gastropoda	Sorbeoconcha	Cochliopidae		S
Arthropoda	Insecta	Diptera	Chironomidae		GC
Mollusca	Bivalvia	Unionida	Hyriidae		FC
Arthropoda	Insecta	Coleoptera	Hydrophilidae		P
Arthropoda	Insecta	Hemiptera	Belostomatidae	*Horvathinia*	P
Mollusca	Bivalvia	Unionida	Mycetopodidae	*Anodontities*	FC
Mollusca	Bivalvia	Veneroida	Corbiculidae	*Corbicula*	FC
Arthropoda	Insecta	Hemiptera	Corixidae		S
Mollusca	Gastropoda	Basommatophora	Ancylini		S
Arthropoda	Malacostraca	Amphipoda	Hyalellidae		GC
Mollusca	Gastropoda	Basommatophora	Planorbidae		S
Mollusca	Gastropoda	Basommatophora	Physidae		S
Arthropoda	Malacostraca	Decapoda	Palaemonidae		O

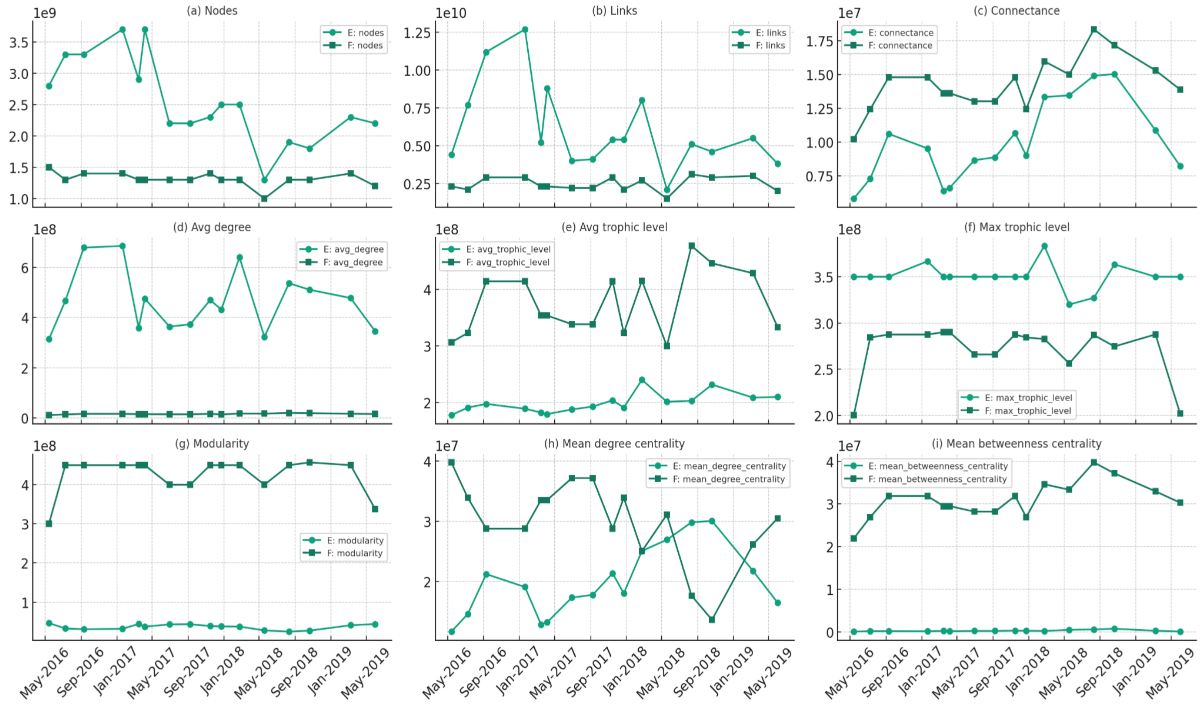

Figure S2. Temporal variation of food web metrics for structural (E, light green lines with circles) and functional networks (F, dark green lines with squares). Each panel (A–I) corresponds to a specific network metric (as indicated), illustrating differences between taxonomic and functional representations of the lagoon food web over time.

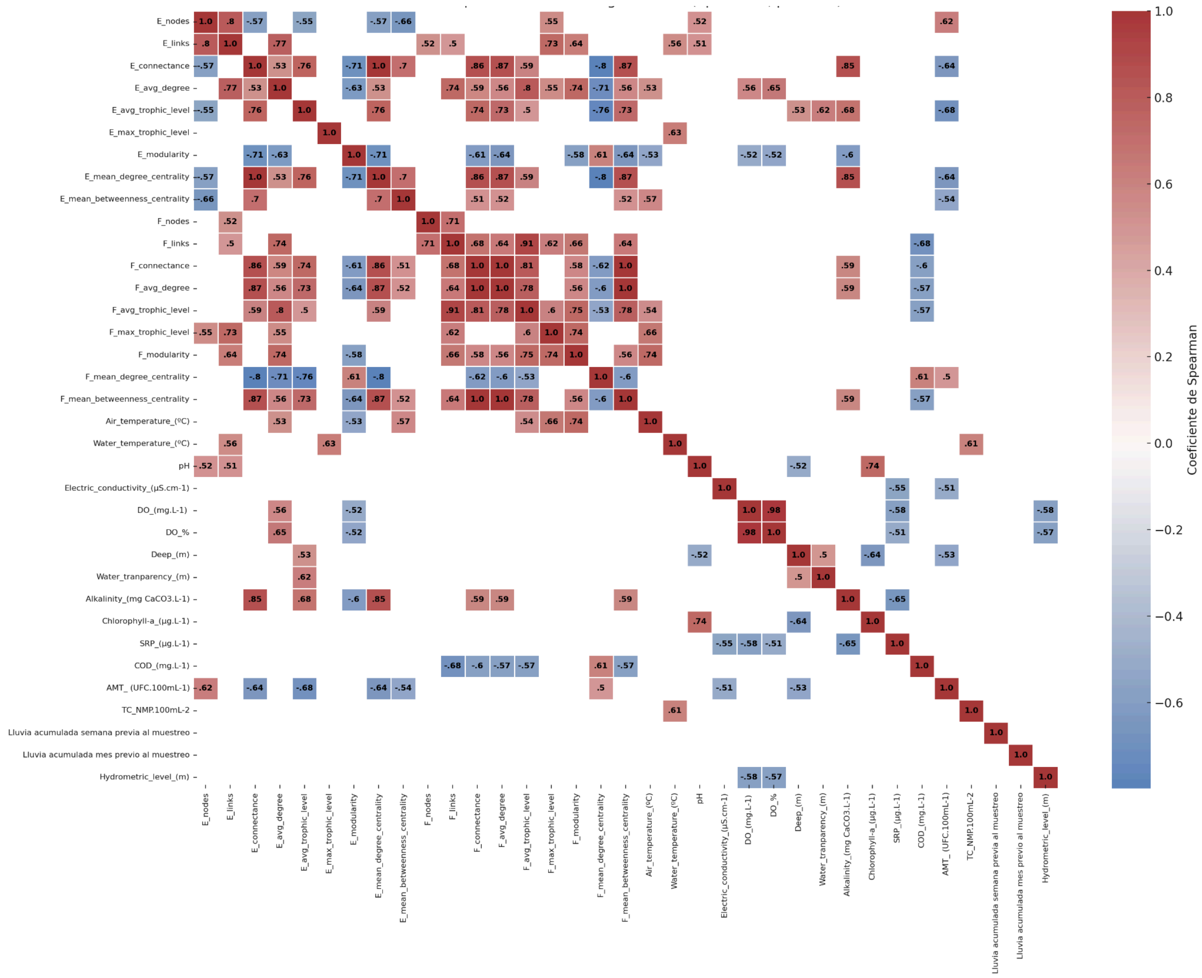

Figure S3. Heatmap of significant correlations (Spearman's coefficient, $p < 0.05$) between network metrics (E: structural food web; F: functional food web) and environmental variables measured in the lagoon. The cells show the correlation coefficients, with the color scale ranging from negative (blue) to positive (red).

References

Crettaz-Minaglia, M. C., Gianello, D., Roldán, C., Chaves, E., Aguer, I., & Juárez, R. A. (2018). Estructura del ensamblaje de macroinvertebrados bentónicos de una laguna urbana de la Pampa Mesopotámica. *Intropica*, 112-121.

Crettaz-Minaglia, M. C., & Gianello, D. (2023). Bacteriological and eutrophication risk assessment of an Argentinian temperate shallow urban lagoon. Environmental Science and Pollution Research, 30(40), 93014-93029.

Cummins, K. W., Merritt, R. W., & Andrade, P. C. (2005). The use of invertebrate functional groups to characterize ecosystem attributes in selected streams and rivers in south Brazil. Studies on Neotropical Fauna and Environment, 40(1), 69-89.

Gerroth, J. (2003). Conjugating Green Algae and Desmids. En Freshwater Algae of North America Ecology and Classification (pp. 353-385). Wehr JD and Sheath RG (eds.)

Gianello, D. 2016. Diagnóstico y gestión ambiental en la Laguna del Parque Unzué (Gualeguaychú, Entre Ríos) con énfasis en el estudio de la diversidad de fitoplancton. Thesis of Grade. Facultad de Ciencia y Tecnología, Universidad Autónoma de Entre Ríos, Argentina.

Gianello, D., Aguer, I., & Crettaz Minaglia, M. C. (2018). Aportes al conocimiento del fitoplancton de la laguna del Parque Unzué (Gualeguaychú, Argentina). *Natura Neotropicalis*, 2.

Gianello, D., Ávila-Hernández, E., Aguer, I., & Crettaz-Minaglia, M. C. (2019). Water quality assessment of a temperate urban lagoon using physico-chemical and biological indicators. *SN Applied Sciences*, *1*(5), 470.

Kingston, J.C. (2003). Araphid and Monoraphid Diatoms. In: *Freshwater Algae of North America Ecology and Classification* (Wehr, J.D., Sheath, R.G. Eds) 595–636.

Kociolek, J.P. & Spaulding, S.A. (2003a). Eunotiod and Asimetrical Naviculoids Diatoms. En: Freshwater Algae of North America Ecology and Classification (Wehr JD, Sheath RG, Eds.) 665–667.

Kociolek, J.P. & Spaulding, S.A. (2003b). Symmetrical Naviculoid Diatoms. In: Freshwater Algae of North America Ecology and Classification (Wehr JD, Sheath RG, Eds.), 637–653.

Komárek, J., & Komarkova, J. (2003). Filamentous Cyanobacteria. In: Freshwater Algae of North America Ecology and Classification (Wehr JD, Sheath RG, eds.) 117–195.

Padisák J, Crossetti LO, Naselli-Flores L (2009) Use and misuse in the application of the phytoplankton functional classification: a critical review with updates. Hydrobiologia 621(1):1–19. https:// doi.org/10.1007/s10750-008-9645-0.

Polazzo, F., Marina, T. I., Crettaz-Minaglia, M., Rico, A. (2022). Food web rewiring drives long-term compositional differences and late-disturbance interactions at the community level. Proceedings of the National Academy of Sciences, 119(17), e2117364119.

Reynolds CS, Huszar V, Kruk C, Naselli-Flores L, Melo S (2002) Towards a functional classification of the freshwater phytoplankton. J Plankton Res 24(5):417–428. https://doi.org/10.1093/plank t/24.5.417

Rodríguez-Barrios, J., Ospina-Tórres, R., & Turizo-Correa, R. (2011). Grupos funcionales alimentarios de macroinvertebrados acuáticos en el río Gaira, Colombia. Revista de Biología Tropical, 59(4), 1537-1552.

Shubert, L.E. (2003). Nonmotile coccoid and Colonial Green Algae. In *Freshwater Algae of North America Ecology and Classification*, Wehr, J.D. y Sheath R.G., Eds, pp. 2053–2309. Elsevier Science, California, USA.

Stoermer, E.F. & Julius, M.L. (2003). Centric diatoms. In *Freshwater Algae of North America* (pp. 559-594). Academic Press.